Topology Optimization of Planar Elastic Energy-Storing Structures in a Fixed Area

Lab III – Project Proposal

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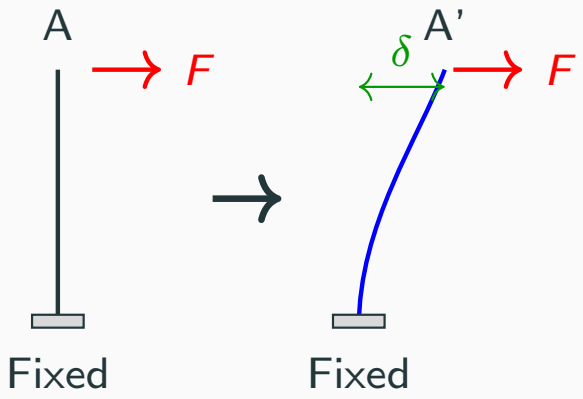
Background: Elastic Energy Storage

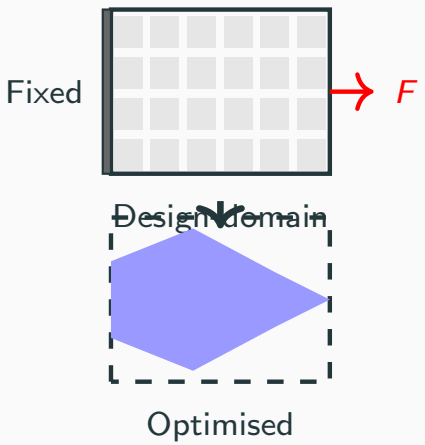
- Planar elastic structures store energy through reversible deformation under load.
- Applications: compliant mechanisms, MEMS actuators, soft robotic grippers, mechanical springs [1].
- Key metric: **energy storage efficiency**

$$\eta = \frac{E_{\text{out}}}{E_{\text{in}}} = \frac{\int F_{\text{out}}\ \mathrm{d}x}{\int F_{\text{in}}\ \mathrm{d}x}$$



Topology Optimization for Structures

- **Topology optimization**: find optimal material layout in a fixed domain [2].
- SIMP method: density $\rho_e \in [0, 1]$ per element [3].
- Applied in aerospace, automotive, biomedical [4].
- **Gap:** few studies experimentally validate topology-optimised planar energy-storing structures [5], [6].



Project Goals

1. **Define objective function** for elastic energy storage performance within a fixed planar footprint.
2. **Generate candidate structures** via:
   - Manual baseline designs (straight, folded, serpentine, curved beams)
   - Computational topology optimisation (SIMP method)
3. **Fabricate and test** all candidates using 3D-printed PLA samples.
4. **Compare** force–displacement behaviour, energy storage E_{in}, energy release E_{out}, and efficiency η across topologies.
5. **Analyse** how topology affects load paths, deformation modes, and stiffness distribution.

Hypotheses

- **H1:** Topologies with distributed deformation (e.g. serpentine or curved structures) will achieve higher energy storage efficiency than straight beam structures.
  - *Rationale:* More uniform deformation reduces stress concentration and energy loss.
- **H2:** Topology-optimized structures will outperform manually designed baselines in terms of energy efficiency and performance stability.
  - *Rationale:* Optimization explores non-intuitive designs that better balance stiffness and compliance.

Theoretical Basis

- **SIMP method [7]** [3], [8]: each element e carries a pseudo-density $\rho_e \in [0, 1]$. Effective stiffness:

$$E_e = \rho_e^p \, E_0, \qquad p \geq 3$$

- **Objective**: minimise structural compliance (maximise stiffness) subject to a volume fraction constraint f:

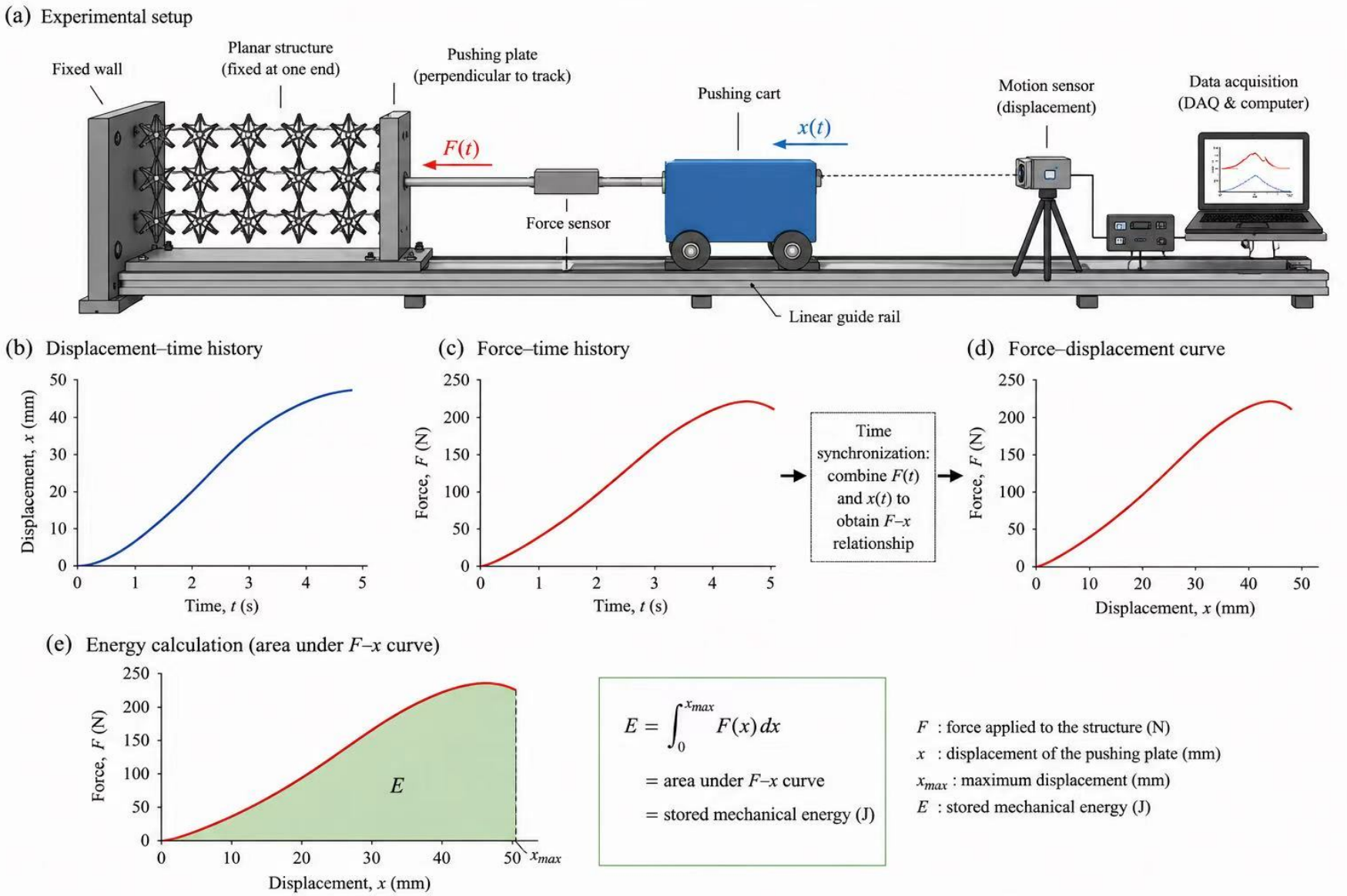
$$\min_{\boldsymbol{\rho}} \; \mathbf{F}^\top \mathbf{U} \quad \text{s.t.} \quad \frac{\sum \rho_e v_e}{V_0} \leq f, \quad \mathbf{K}(\boldsymbol{\rho}) \, \mathbf{U} = \mathbf{F}$$

- **Elastic energy stored**:

$$E_{\text{in}} = \int_0^{\delta_{\text{max}}} F(\delta) \, \mathrm{d}\delta$$

  measured from force–displacement curve.
- Efficiency $\eta = E_{\text{out}} / E_{\text{in}}$ captures hysteresis and dissipation losses.

Experimental Setup — Overview



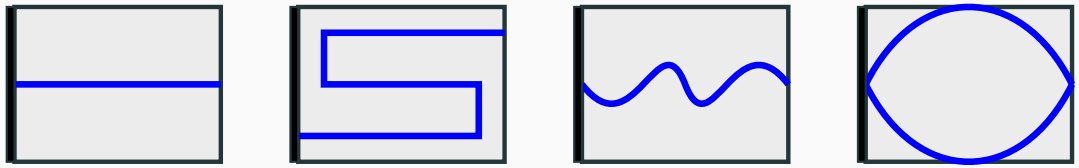
Experimental Setup — Overview

- **Workflow:** Quasi-static compression $\rightarrow$ Time-synced $F(t)\&x(t)$ acquisition $\rightarrow$ $F - x$ relationship transformation for energy evaluation.
- **Loading Method:** Quasi-static manual pushing.
- **Data Acquisition:**
  - Displacement $x(t)$ recorded by Motion Sensor.
  - Applied Force $F(t)$ recorded by Force Sensor.
- **Stability:** High linearity in $x(t)$ indicates steady manual loading.

Apparatus & Materials

- **Platform:** Hand wheel + lead screw compression rig (custom-built)
- **Force measurement:** Digital gauge, 0–50 N, 0.1 N resolution
- **Displacement:** Digital calliper, ± 0.01 mm
- **Material:** PLA filament (1.75 mm), FDM 3D printing
- **Software:** GNU Octave, FreeCAD, Python
- **Environment:** Room temperature ($\sim 25\,^{\circ}\mathrm{C}$)
- **Material:** 3D printed PETG (1.75 mm); chosen for its superior toughness and interlayer adhesion compared to PLA, ensuring structural integrity during large deformations.

Candidate Structures



(a) Straight (b) Folded (c) Serpentine (d) Curved

Four manual baselines + 1–2 topology-optimised candidates. All share identical footprint, thickness, and boundary conditions.

- **Design space:** 4 manual baselines vs. 1–2 topology-optimized candidates.
- **Constraints:** Identical footprint ($W \times L$), thickness, and PETG material.
- **Goal:** To evaluate the influence of geometric topology on energy storage E.

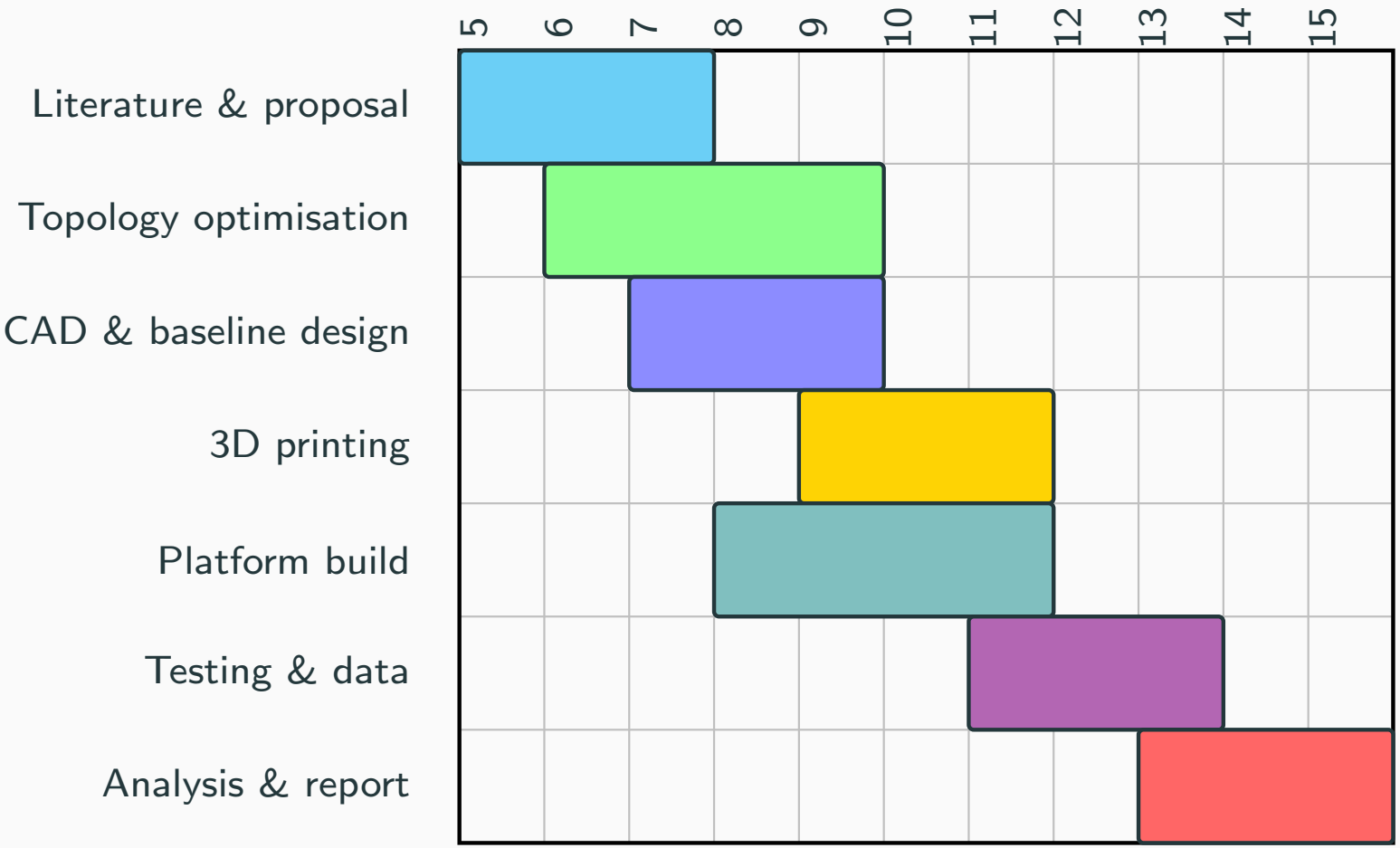
Analysis Plan

1. **Data acquisition:** Record force F and displacement δ at 0.1 mm increments during quasi-static compression. Repeat each topology ≥ 3 times.
2. **Preprocessing:** Zero-offset correction; apply force sensor calibration curve; discard first cycle (pre-conditioning).
3. **Feature extraction:**
   - Stiffness: slope of linear region of F–δ curve.
   - E_{in}: area under loading curve (trapezoidal integration).
   - E_{out}: area under unloading curve.
   - $\eta = E_{\text{out}}/E_{\text{in}}$.
4. **Statistical treatment:** Mean $\pm$ standard deviation across trials; compare topologies using bar charts with error bars.
5. **Validation:** Compare experimental stiffness with FEA prediction (if time permits, COMSOL cross-check).

Safety & Compliance

- **Hazards identified:**
  - 3D printer: hot nozzle ($\sim 200\,^{\circ}\mathrm{C}$), moving parts, minor fume emission from PLA.
  - Compression testing: spring-back of samples, pinch risk from lead screw mechanism.
- **Mitigation measures:**
  - Operate printer in ventilated area; do not touch nozzle during operation.
  - Wear safety glasses during compression tests.
  - Use hand wheel with controlled speed; never exceed elastic limit of samples.
- **Regulatory status:** No chemicals, radiation, or biological agents involved. Not regulated under Malaysian Occupational Safety and Health Act 1994.
- **Waste disposal:** PLA is biodegradable; failed prints collected in general recycling bin.
- **Emergency contacts:** Lab supervisor; campus security (ext. 8888).

Project Timeline



Budget

Item	Qty	Unit (MYR)	Total (MYR)
Equipment			
Digital force gauge (0–50 N)	1	45	45
Lead screw + guide rail set	1	35	35
Hand wheel	1	10	10
Materials & Consumables			
PLA filament (1 kg spool)	1	40	40
Acrylic base plate	1	15	15
Bolts, nuts, clamps	–	–	20
Software & Licences			
Open-source (Octave, FreeCAD, Python)	–	–	0
Miscellaneous			
Spare filament & reprints	–	–	20
Total			**185**

Requested from Physics Dept. student project fund (cap: MYR 500). 3D printer access via university makerspace.

References i

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[4] J. Liu et al., "Current and future trends in topology optimization for additive manufacturing," *Structural and Multidisciplinary Optimization*, vol. 57, no. 6, pp. 2457–2483, 2018. DOI: 10.1007/s00158-018-1994-3

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[7] Y.-X. Toh, Y.-T. Chin, E. Q. Wu, and Y.-K. Lim, "Properties of the magnetic universe with positive cosmological constant," *Classical and Quantum Gravity*, vol. 42, no. 17, p. 177 001, 2025, ISSN: 1361-6382. DOI: 10.1088/1361-6382/adfc1d [Online]. Available: http://dx.doi.org/10.1088/1361-6382/adfc1d

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[8] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund, "Efficient topology optimization in MATLAB using 88 lines of code," *Structural and Multidisciplinary Optimization*, vol. 43, no. 1, pp. 1–16, 2011. DOI: 10.1007/s00158-010-0594-7